

K MEANS CLUSTERING

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Abstract— This is an experiment where we can see the effect of k means clustering from a given datasets with MATLAB scripting

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INTRODUCTION

In this experiment, we designed a K means clustering from a given dataset where we draw three different clustering classes from a given 150 data set.

OBJECTIVE

The basic step of k-means clustering is simple. In the beginning we determine number of cluster K and we assume the centroid or center of these clusters. We can take any random objects as the initial centroids or the first K objects in sequence can also serve as the initial centroids.

Then the K means algorithm will do the three steps below until convergence Iterate until stable (= no object move group):

1. Determine the centroid coordinate
2. Determine the distance of each object to the centroids
3. Group the object based on minimum distance

TASK 1: PLOTING SAMPLES & DETERMINGING THE PRIMARY CLUSTERING

Our dataset is given by a text file names irisdataset.txt. Now, we have to plot these data using MatLab's Figure and scripting. We can do this by the following code given below.

```
x=zeros(150,4);  
  
myfile=fopen('irisdataset.txt','r');  
  
x=fscanf(myfile,'%f\t%f\t%f\t%f',size(x));
```

```
figure,hold on;  
%plot(x(:,1), x(:,2),x(:,3),x(:,4), 'bs');  
grid on;  
hold off;  
  
randomRow = randi([1,150]);  
c1 = [x(randomRow,1) x(randomRow,2)  
x(randomRow,2) x(randomRow,4)];  
  
randomRow = randi([1,150]);  
c2 = [x(randomRow,1) x(randomRow,2)  
x(randomRow,2) x(randomRow,4)];  
  
randomRow = randi([1,150]);  
c3 = [x(randomRow,1) x(randomRow,2)  
x(randomRow,2) x(randomRow,4)];
```

We can see that we chose a random variable to choose 3 clustering groups named c1, c2, c3 from given data set X.

TASK 2: DETERMINGING THE DISTANCE FROM CENTROIDS

Here we are determining the distance between our chosen centroids and our dataset named x to find the distance matrix explained later.

```
d1 = zeros(1,150);  
d2 = d1;  
d3 = d1;  
for i = 1:150  
    d1(1,i) = sqrt((c1(:,1) - x(i,1)) *  
    (c1(:,1) - x(i,1)) + (c1(:,2) - x(i,2)) *  
    (c1(:,2) - x(i,2)) + (c1(:,1) - x(i,3)) *  
    (c1(:,1) - x(i,3)) + (c1(:,2) - x(i,4)) *  
    (c1(:,2) - x(i,4))) );  
    d2(1,i) = sqrt((c2(:,1) - x(i,1)) *  
    (c2(:,1) - x(i,1)) + (c2(:,2) - x(i,2)) *  
    (c2(:,2) - x(i,2)) + (c2(:,1) - x(i,3)) *  
    (c2(:,1) - x(i,3)) + (c2(:,2) - x(i,4)) *  
    (c2(:,2) - x(i,4))) );  
    d3(1,i) = sqrt((c3(:,1) - x(i,1)) *  
    (c3(:,1) - x(i,1)) + (c3(:,2) - x(i,2)) *  
    (c3(:,2) - x(i,2)) + (c3(:,1) - x(i,3)) *  
    (c3(:,1) - x(i,3)) + (c3(:,2) - x(i,4)) *  
    (c3(:,2) - x(i,4))) );  
end
```

```

(c2(:,1) - x(i,3)) + (c2(:,2) - x(i,4)) *
(c2(:,2) - x(i,4))) ;
    d3(1,i) = sqrt((c3(:,1) - x(i,1)) *
(c3(:,1) - x(i,1)) + (c3(:,2) - x(i,2)) *
(c3(:,2) - x(i,2)) + (c3(:,1) - x(i,3)) *
(c3(:,1) - x(i,3)) + (c3(:,2) - x(i,4)) *
(c3(:,2) - x(i,4))) ;
end;
d = [d1; d2; d3];

```

We followed the normal distance measurement (square root of $(x_1-x_2)^2 + (y_1-y_2)^2$) formula to find the distance.

TASK 3: GROUPING TOGETHER

We assign each object based on the minimum distance. The element of Group matrix below is 1 if and only if the object is assigned to that group.

```

g = d;
for j = 1:150
    min = 50000;
    index = 0;
    for i = 1:3
        if(d(i,j) < min)
            min = d(i,j);
            index = i;
        end;
    end;
    for i = 1:3
        g(i,j) = 0;
    end;
    g(index,j) = 1;
end;

```

TASK 4: CLUSTERING UNTIL GROUPS ARE IDENTICAL

Now we run clustering loop to find out the classes assigned with the 3 groups.

```

%CLUSTERING LOOP
for w=1:20
    tempG = g;
    for j = 1:150
        for i = 1:3
            if(g(i,j)==1 && i==1)
                newX1 = (x(j,1) + c1(1,1))/2;
                newX2 = (x(j,2) + c1(1,2))/2;
                newX3 = (x(j,3) + c1(1,3))/2;
                newX4 = (x(j,4) + c1(1,4))/2;
                c1 = [newX1 newX2 newX3 newX4];
            elseif(g(i,j)==1 && i==2)
                newX1 = (x(j,1) + c2(1,1))/2;
                newX2 = (x(j,2) + c2(1,2))/2;
                newX3 = (x(j,3) + c2(1,3))/2;
                newX4 = (x(j,4) + c2(1,4))/2;
                c2 = [newX1 newX2 newX3 newX4];
            elseif(g(i,j)==1 && i==3)
                newX1 = (x(j,1) + c3(1,1))/2;
                newX2 = (x(j,2) + c3(1,2))/2;
                newX3 = (x(j,3) + c3(1,3))/2;
                newX4 = (x(j,4) + c3(1,4))/2;
                c3 = [newX1 newX2 newX3 newX4];
            end;
        end;
    end;
    d1 = zeros(1,150);
    d2 = d1;
    d3 = d1;
    for i = 1:150
        d1(1,i) = sqrt((c1(:,1) - x(i,1)) *
(c1(:,1) - x(i,1)) + (c1(:,2) - x(i,2)) *
(c1(:,2) - x(i,2)) + (c1(:,1) - x(i,3)) *
(c1(:,1) - x(i,3)) + (c1(:,2) - x(i,4)) *
(c1(:,2) - x(i,4))) ;
        d2(1,i) = sqrt((c2(:,1) - x(i,1)) *
(c2(:,1) - x(i,1)) + (c2(:,2) - x(i,2)) *
(c2(:,2) - x(i,2)) + (c2(:,1) - x(i,3)) *
(c2(:,1) - x(i,3)) + (c2(:,2) - x(i,4)) *
(c2(:,2) - x(i,4))) ;
        d3(1,i) = sqrt((c3(:,1) - x(i,1)) *
(c3(:,1) - x(i,1)) + (c3(:,2) - x(i,2)) *
(c3(:,2) - x(i,2)) + (c3(:,1) - x(i,3)) *
(c3(:,1) - x(i,3)) + (c3(:,2) - x(i,4)) *
(c3(:,2) - x(i,4))) ;
    end;
    d = [d1; d2; d3];
    g = d;
    for j = 1:150
        min = 50000;
        index = 0;
        for i = 1:3
            if(d(i,j) < min)
                min = d(i,j);
                index = i;
            end;
        end;
        %display(index);
        for i = 1:3
            g(i,j) = 0;
        end;
        g(index,j) = 1;
    end;
end;

```

```

if(tempG==g)
    break;
else
    tempG=g;
end;
end;
display(w);

```

Here 'w' is the number of loops needed to find the identical group matrix that we are looking for.

TASK 5: PLOTTING THEM TOGETHER

Now we need to plot the clusters into the grid to find out the result. The code for plotting the values are given below:

```

hold on;
plot(c1(:,1), c1(:,2),c1(:,3),c1(:,4),
'bs*');
hold off;
hold on;
plot(c2(:,1), c2(:,2),c2(:,3),c2(:,4),
'r*');
hold off;
hold on;
plot(c3(:,1), c3(:,2),c3(:,3),c3(:,4),
'k*');
hold off;

```

```

counterG1=0;
counterG2=0;
counterG3=0;
for j = 1:150
    for i = 1:3
        if(g(i,j)==1 && i==1)
            counterG1=counterG1+1;
            hold on; plot(x(j,1),
x(j,2),x(j,3),x(j,4), 'bs'); hold off;
        elseif(g(i,j)==1 && i==2)
            counterG2=counterG2+1;
            hold on; plot(x(j,1),
x(j,2),x(j,3),x(j,4), 'rs'); hold off;
        elseif(g(i,j)==1 && i==3)
            counterG3=counterG3+1;
            hold on; plot(x(j,1),
x(j,2),x(j,3),x(j,4), 'ks'); hold off;
        end;
    end;
end;
display(counterG1);
display(counterG2);
display(counterG3);

```

Here, 'counterG1', 'counterG2', 'counterG3' variable denotes that how many samples represent each cluster group.

The final plot we have here is:

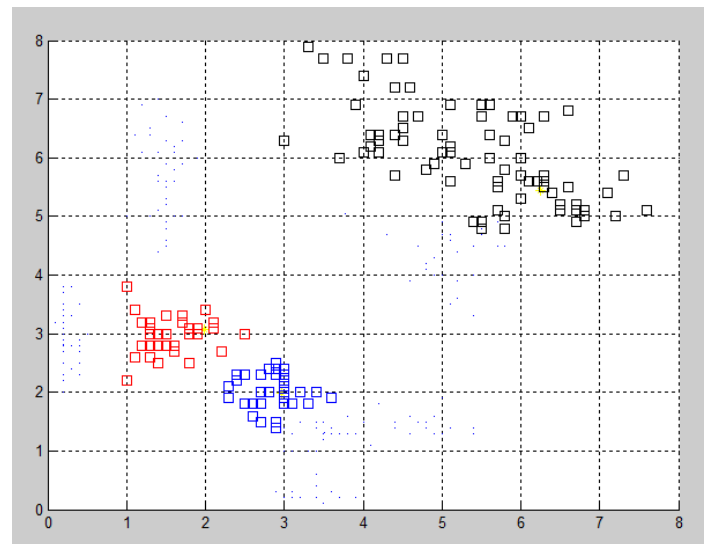


Figure01 – Clustering Groups

RESULT ANALYSIS

From the figure -1, we can see that we have found three groups colored red, blue, and black from a randomly chosen centroids and tuning it to perfection until we found the identical set of matrix to represent the clusters.

REFERENCES

[1] Pattern Classification (2nd Edition) by R. O. Duda, P. E. Hart and D. Stork, Wiley 2002